



Multi-scale simulation of solidification behavior and microstructure evolution during vacuum electron beam welding of Al-Cu alloy

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ABSTRACT

A multi-scale model is established to study the complex melting and solidification behaviors during the electron beam welding of an Al-Cu alloy. The model combines a macroscopic thermal-fluid algorithm for the molten pool behavior and a microscopic cellular automaton algorithm for the microstructure evolution. The solidification parameters, including the thermal gradient and cooling rate, build a bridge between the macroscopic and microscopic simulations. The model is validated by comparing the experimental fusion profile results, microstructure, morphology, grain size and element distribution that result from different welding parameters. On the macroscopic scale, with an increase in beam current, the molten pool fluctuation intensity increases, the thermal gradient increases and the cooling rate decreases. On the microscopic scale, the grain growth process includes nucleation, growth and competitive growth. After solidification, the concentration of Cu on the grain boundaries is larger than that inside the grains. With increasing beam current, the grain size tends to increase, and the segregation ratio decreases. The power-law function between the grain size and cooling rate can be summarized as $d \propto CR^{-0.55}$, where d is the grain size and CR is the cooling rate. Moreover, the segregation ratio first increases and then decreases slowly with decreasing solidification time. We expect this study to provide a deeper understanding of how to control weld formation and microstructure and improve confidence in optimizing the actual welding process.

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1. Introduction

2219 aluminum alloy has been widely applied in the aeronautical and automotive industries to produce lightweight structures [1,2]. Due to its characteristics of high energy density and high productive efficiency, electron beam welding (EBW) is particularly suitable for welding dissimilar metals and materials with high thermal conductivity, such as aluminum alloys [3,4]. The microstructure evolution during the solidification process of the molten pool is the critical factor affecting the mechanical properties of welded joints [5–7]. An improved understanding of the solidification behavior enables the optimization of the EBW process parameters and improves the quality of the weld joints [8].

Recently, scholars adopted experimental methods, such as the use of X-ray imaging [9], high-speed charge-coupled device (CCD) filming [10], thermocouples [11] and in situ observation [12], to monitor the dynamic welding process. These investigations can be used to provide valuable insights into the melting and solidification process in a molten pool. However, only low-quality images

can be obtained by experimental observations. Additionally, the experimental measurements do not provide an accurate and quantitative description of the thermal gradient and cooling rate at the solidification interface due to the rapid cooling of the melt. Therefore, there is an urgent need for a new method to visualize the dynamic welding process and quantify the temperature field in molten pools. After welding, the microstructure information of the weld seam can be studied by many material characterization methods, such as optical microscopy (OM) and scanning electron microscopy (SEM) [13]. Although the microstructure evolution mechanism can be deduced according to solidification theory and characterization results, the evolution of the concentration field cannot be reproduced during the solidification process. Therefore, it is of great significance to find a method that can reproduce microstructure evolution and quantitatively predict grain size.

With the development of computational materials science, the dynamic welding process can be visualized and analyzed quantitatively by a multi-scale model. Yu et al. [14] and Geng et al. [15] calculated the welding temperature field by the finite element method and simulated the columnar grain growth by the phase-field (PF) model during the solidification process of the molten pool. They discussed the growth mechanism of dendrites in the

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molten pool. Wang et al. [16] and Xiao et al. [17] established a macro-micro model to predict the primary dendritic arm spacing (PDAS) and microsegregation ratio after solidification without considering the effect of columnar grain growth orientation and random nucleation. An expression of the relationship between the solidification parameters and PDAS was proposed. Researchers [18–20] predicted the evolution of grain structures in molten pools by a cellular automaton-finite element (CA-FE) model. Their simulation results are in good agreement with the experimental data. The existing studies have demonstrated the ability of the numerical method to predict microstructure evolution. However, most of the investigations on the multi-scale modeling of the welding process only considered the grain growth rate as a function of temperature and ignored the microsegregation caused by the concentration field distribution. In addition, for the majority of macroscopic models, the effect of molten pool fluctuation on the solidification parameters was also neglected, which may lead to a deviation in the prediction of microstructures. Therefore, the prediction of the microstructure and segregation for different welding parameters requires further investigation.

In this study, a multi-scale CA-FVM model is established to simulate the dynamic fluctuation of the molten pool and microstructure evolution during the EBW process. The most innovative feature of the developed model is a full consideration of electron beam-keyhole interactions, multi-force driven molten pool behavior, grain growth and element segregation. The keyhole behavior and solidification parameters (thermal gradient and cooling rate) related to welding quality under different welding parameters are discussed briefly. Then, we focus on the influences of the solidification parameters on the microstructure morphology, grain size and segregation. The simulated results are compared with those from experiments and the results of other studies. This present work will contribute a high-quality model for predicting the weld seam formation of the EBW process and could be helpful for not only understanding and controlling the microstructure morphology but also optimizing the actual welding process.

2. Experimental details

In this study, a large vacuum KL110 EBW machine was utilized for the welding experiments. The material used in the experiments was an aluminum alloy sheet with a thickness of 20 mm. The base metal was clamped and welded in a vacuum chamber after cleaning under a vacuum of 5×10^{-2} Pa. The accelerating voltage of 60 kV, welding speed of 10 mm/s, working distance of 200 mm and focusing current of 730 mA were selected as the process parameters. Four different cases were conducted: case A with a beam current of 40 mA, case B with a beam current of 50 mA, case C with a beam current of 55 mA and case D with a beam current of 60 mA.

After welding, the welding samples were cut along the middle region of the joints. Standard grinding and mechanical polishing were used during the preparation of the observed specimens. Keller's reagent was adopted as an etchant for material characterization. The grain structures and sizes were visualized and measured on a Quanta 200FEG equipped with an electron backscattered diffraction (EBSD) system. The element concentrations in the dendrite core and inter-dendritic region were acquired through energy dispersive spectroscopy (EDS).

3. Models and methodology

Fig. 1(a) introduces the basic ideas and the outline of this paper and finally establishes the research approach. First, a thermal-fluid model that reproduces the typical keyhole effect during EBW is developed to present the molten pool fluctuation process at the

macroscale. Second, the thermal gradient and the cooling rate obtained from the above model are taken as the initial conditions for the solidification process for the microstructure simulation. Based on these solidification parameters, the microstructure morphology, grain size and segregation can be predicted using a cellular automaton method. Finally, some of the simulated results are verified quantitatively by different EBW experiments. By comparing the simulated results with the experimental results, the accuracy of the macroscopic thermal-fluid model can be inferred from the fusion profile [21], and the accuracy of the microscopic cellular automaton model can be inferred from the microstructure morphology, grain size and element distribution [16].

3.1. Macroscopic thermal-fluid model for EBW

A three-dimensional numerical model for heat transfer and fluid flow in a dynamic molten pool has been established. A detailed description of the model can be found in our previous studies [22], and the governing equations and boundary conditions are briefly introduced here for the sake of completeness. Fig. 1(b) shows the macroscopic geometric model and computational domain in this study, and its dimensions are 30 mm $\times$ 10 mm $\times$ 22 mm ($x \times y \times z$). A 2 mm region is set on a 20-mm-thick base metal in the z -direction to track the molten pool and keyhole surfaces.

By assuming that the liquid metal is a laminar, incompressible fluid with Newtonian viscosity, the continuity equation, momentum equation and energy equation are solved to describe the melt flow in the molten pool, which are presented as follows [23]:

$$\nabla \cdot \vec{U} = 0 \quad (1)$$

$$\rho \left(\frac{\partial \vec{U}}{\partial t} + (\vec{U} \cdot \nabla) \vec{U} \right) = \nabla \cdot (\mu \nabla \vec{U}) - \nabla p - \rho \vec{g} \beta (T - T_{ref}) \quad (2)$$

$$\rho C_p \left(\frac{\partial T}{\partial t} + (\vec{U} \cdot \nabla) T \right) = \nabla \cdot (k \nabla T) \quad (3)$$

where $\vec{U}$, ρ , μ , p , $\vec{g}$, β , T , T_{ref} , C_p and k are the flow velocity, density, kinetic viscous coefficient, pressure, gravitational vector, thermal expansion coefficient, temperature, reference temperature, specific heat capacity and thermal conductivity, respectively. The molten pool surface and the keyhole wall are tracked by the volume of fluid (VOF) algorithm [24]:

$$\frac{\partial F}{\partial t} + \nabla \cdot (\vec{U} \cdot F) = 0 \quad (4)$$

where F is the fluid volume fraction. If $F=0$, then the cell contains only a vacuum; if $F=1$, then the cell is full of liquid metal; and if $0 < F < 1$, then the cell is at the vacuum and liquid metal interface.

The boundary conditions, which play an important role in numerical simulation, can also be seen in Fig. 1(b). The front surface is regarded as a symmetric boundary condition for improving computational efficiency. The top surface is set to the outflow boundary condition, which is extrapolated from within the domain and has no impact on the upstream flow. Other surfaces are set to the wall boundary condition, which means that only radiation is considered. The complex phenomena in the molten pool are the result of thermal-mechanical coupling [25]. During the welding process, the main thermal boundary conditions, including the electron beam energy q_{ebw} , radiation q_{rad} and evaporation q_{evap} , are modeled by the following formula [22]:

$$K \frac{\partial T}{\partial n} = q_{ebw} - q_{rad} - q_{evap} = q_{ebw} - \varepsilon \sigma (T^4 - T_{ref}^4) - m_v H_v \quad (5)$$

where ε , σ , m_v and H_v denote the surface radiation emissivity, Stefan-Boltzmann constant, evaporation mass and latent heat of

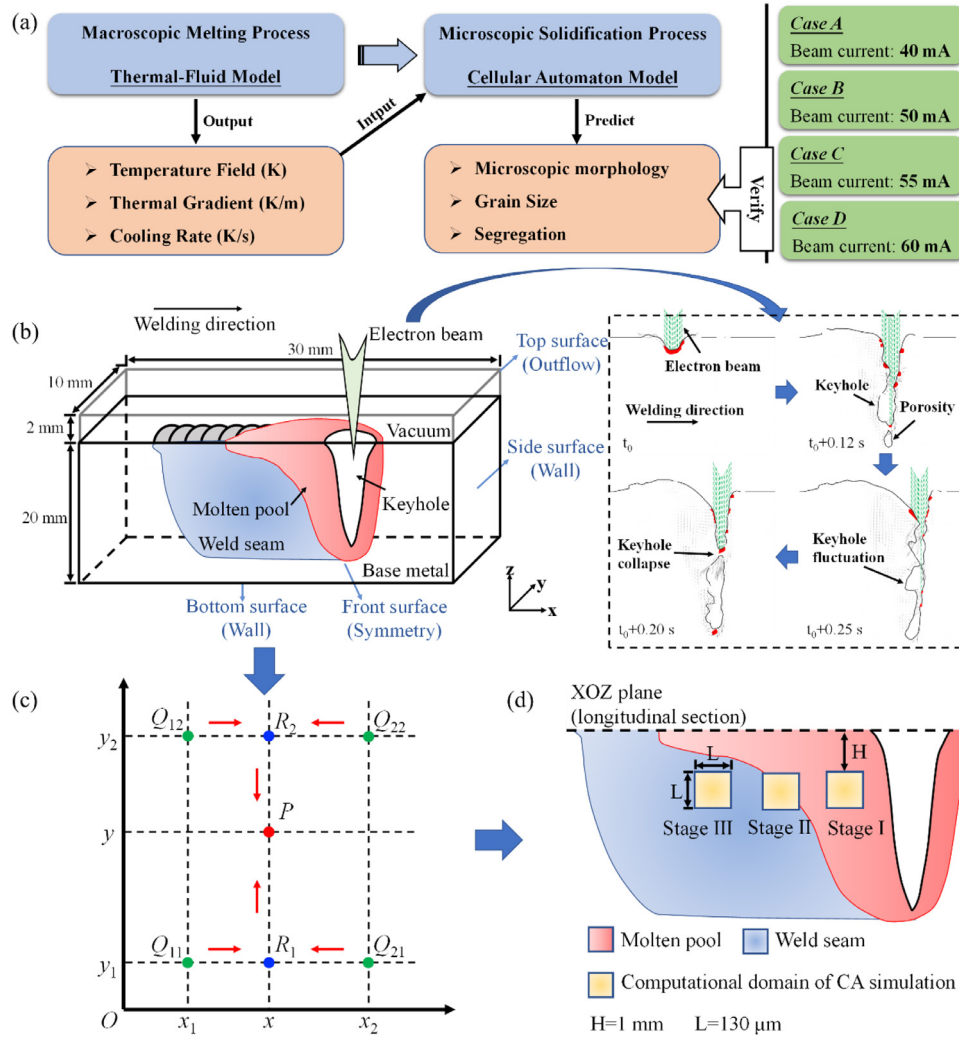


Fig. 1. Schematic diagram of the multi-scale model used for simulating melting and solidification processes during the EBW process: (a) the methodology and research approach, (b) the computational domain and boundary conditions of the macroscopic thermal-fluid model, (c) the principle of bilinear interpolation, which is used to transform the temperature field from macro to micro, and (d) the computational domain of the microscopic CA model. The inset in Fig. 1(b) is a schematic diagram of the interaction between the electron beam and keyhole wall.

evaporation, respectively. To reflect the physical nature of the interaction between the electron beam and keyhole wall, an innovative dynamic surface heat source model is established to accurately apply the body-flux heat source model to the material with rapidly evolving free surfaces. The heating method applied on the keyhole wall at different times can be reflected in the inset on the right of Fig. 1(b). The horizontal power density distribution of the electron beam can be supposed to have a Gaussian shape [26]. For this distribution, the power density of the electron beam energy can be described as [27]:

$$q(x, y, z) = \frac{3\eta Q}{\pi r_z^2} \exp\left(-3\frac{x^2 + y^2}{r_z^2}\right) \quad (6)$$

$$r_z = r_0 + a|z_0 - z| \quad (7)$$

where Q and η are the heat input and welding energy efficiency related to the KL110 EBW machine, respectively; r_z indicates the beam radius under different penetration depths; r_0 is the focal radius of the electron beam at the upper surface of the base metal; and a is an attenuation coefficient, which denotes the degree of beam divergence. It is worth noting that the values of η , a and r_0 are provided by the instruction manual for the KL110 EBW machine.

Additionally, the driving forces play a key role in the shape and stability of the molten pool, especially the surface tension and recoil pressure [28]. In this model, the surface tension P_γ and the recoil pressure P_v induced by vaporization can be expressed by the following formulas [29],[30]:

$$P_\gamma = \kappa \gamma = \kappa \left[\gamma_0 + \frac{d\gamma}{dT} (T - T_m) \right] \quad (8)$$

$$P_v = 0.54P_0 \exp\left(H_m \frac{T - T_b}{RT_b}\right) \quad (9)$$

where κ , γ , γ_0 , $\frac{d\gamma}{dT}$ and T_m represent the free surface curvature, surface tension coefficient, surface tension at the liquidus temperature, gradient coefficient for the surface tension and melting point of the processed material, respectively. Additionally, P_0 , H_m , T_b and R represent the ambient pressure, latent heat of fusion, boiling point of the metal and universal gas constant, respectively.

The EBW experiments are conducted inside a vacuum chamber, and an ideal vacuum environment may not be modeled in the VOF model. To avoid the solution becoming nonphysical while solving the momentum equation, the viscosity and density in the vacuum area cannot be zero. Therefore, it is necessary to make an approximation to define a vacuum. The operating pressure is set to zero;

Table 1
Material properties for 2219 aluminum alloy.

Physical property	Value	Physical property	Value
Density (solid)	2700 kg·m ⁻³	Latent heat of fusion	3.87 × 10 ⁵ J·kg ⁻¹
Density (liquid)	2400 kg·m ⁻³	Latent heat of evaporation	1.08 × 10 ⁷ J·kg ⁻¹
Specific heat (solid)	871 J·kg ⁻¹ ·K ⁻¹	Surface tension at 930 K	0.914 N·m ⁻¹
Specific heat (liquid)	1060 J·kg ⁻¹ ·K ⁻¹	Surface tension gradient	-0.00035 N·m ⁻¹ ·K ⁻¹
Thermal conductivity (solid)	238 W·m ⁻¹ ·K ⁻¹	Solidus temperature	820 K
Thermal conductivity (liquid)	100 W·m ⁻¹ ·K ⁻¹	Liquidus temperature	930 K
Boiling point	2730 K	Ambient temperature	300 K
Universal gas constant	8.31 J·mol ⁻¹ ·K ⁻¹	Gravitational acceleration	9.81 J·mol ⁻¹ ·K ⁻¹

density and viscosity are highly decreased to the range with which the converged solution would be obtained [31].

With the above models, the molten pool fluctuation and the keyhole dynamics can be simulated and computed by using ANSYS Fluent 15.0, and the computation time spent on a common desktop with an Intel Core i7-7700k CPU is approximately 150 h. It should be noted that the mesh sensitivity has been checked by performing mesh convergence tests and the minimum mesh size at the fusion zone is 0.125 mm. The description of the mesh convergence test can be found in the Supplementary materials. The time step adopted in this computation is fixed at 1×10^{-5} s. The governing equations with the boundary conditions are solved by a pressure-implicit with splitting of operators (PISOs) pressure-velocity coupling algorithm. In addition, the user-defined function (UDF) is adopted to compile the energy exchange and driving forces in the molten pool. The thermo-physical parameters and the relevant coefficients for the material used in this simulation are summarized in Table 1 [32].

3.2. Microscopic cellular automaton model for the solidification process

A 2D domain with dimensions of $130 \mu\text{m} \times 130 \mu\text{m}$, which is 1 mm away from the base metal surface, is selected as a microscopic computational domain, as illustrated in Fig. 1(d). This location is selected because the weld center is an important and typical region for material characterization. As the electron beam moves forward (along the x-direction), the temperature field moves forward correspondingly through the CA domain. The whole process can be divided into three different stages. In stage I, the base metal is melted by the electron beam energy. When the temperature is lower than the liquidus temperature, the grains begin to nucleate and grow as the temperature field moves forward, as shown in stage II. Finally, the molten pool is completely solidified, and microstructure evolution is over, as shown in stage III. This is equivalent to the stage at which the CA domain gradually experiences melting and solidification processes.

Heterogeneous nucleation is a common phenomenon during the solidification of a molten pool [20]. Thévoz et al. proposed that the increase in the nucleation density n with the nucleation undercooling ΔT can be assumed to be a Gaussian distribution [33]:

$$\frac{dn}{d(\Delta T)} = \frac{n_{\max}}{\sqrt{2\pi}\Delta T_{\sigma}} \exp\left[-\frac{1}{2}\left(\frac{\Delta T - \Delta T_{\text{ave}}}{\Delta T_{\sigma}}\right)^2\right] \quad (10)$$

where $n_{\max}$, ΔT_{σ} and ΔT_{ave} denote the maximum grain density, standard deviation and average value of the nucleation undercooling, respectively. Additionally, a nucleation probability is proposed to determine the characteristics of random nucleation during the solidification process [20]. The nucleation probability of each CA cell can be expressed as [34]:

$$P_n = V_{CA} \int_{\Delta T}^{\Delta T + \delta(\Delta T)} \frac{dn}{d(\Delta T)} d(\Delta T) \quad (11)$$

where V_{CA} is the volume of a CA cell. When the nucleation probability is greater than a random number (0–1), the cellular state changes from the liquid phase to the solid phase. After nucleation, the dendrites begin to grow under the influence of the temperature and solute migration. The calculation process can be divided into three steps, which are introduced below.

First, the solute transport processes in the liquid and solid phases are solved by the following equations [20]:

$$\frac{\partial C_L}{\partial t} = D_L \left(\frac{\partial^2 C_L}{\partial x^2} + \frac{\partial^2 C_L}{\partial y^2} \right) + C_L(1 - k_0) \frac{\partial f_s}{\partial t} \quad (12)$$

$$\frac{\partial C_S}{\partial t} = D_S \left(\frac{\partial^2 C_S}{\partial x^2} + \frac{\partial^2 C_S}{\partial y^2} \right) \quad (13)$$

where C_L and D_L represent the solute concentration and diffusion coefficient in the liquid phase, respectively; C_S and D_S represent the solute concentration and diffusion coefficient in the solid phase, respectively; and k_0 and f_s represent the partition coefficient and solid fraction, respectively.

Second, the growth kinetics at the solid-liquid interface are mainly determined by solute redistribution, which can be described as [35]:

$$V_n C_L^* (1 - k_0) = (D_S \cdot \nabla C_S - D_L \cdot \nabla C_L) \vec{n} \quad (14)$$

where V_n and $\vec{n}$ represent the growth rate and normal unit vector from the solid-liquid interface, respectively. The solute concentrations around the solid-liquid interface are recorded as C_S^* and C_L^* , which can be expressed as [35]:

$$C_L^* = C_0 + \frac{1}{m_l} (T^* - T_l + \Gamma \kappa f(\varphi, \theta)) \quad (15)$$

$$C_S^* = k_0 C_L^* \quad (16)$$

where C_0 represents the equilibrium concentration in the initial liquid phase; m_l represents the liquidus slope, which can be acquired from the alloy phase diagram; and T^* and T_l indicate the local temperature around the solid-liquid interface and liquidus temperature, respectively. The expression $\Delta T = T^* - T_l$ represents the undercooling, which is a necessary condition for solidification and crystallization [36]. In addition, Γ represents the Gibbs-Thomson coefficient, and κ represents the mean curvature radius, which can be expressed as [37]:

$$\kappa = \frac{2\partial_x f_s \cdot \partial_y f_s \cdot \partial_{xy} f_s - (\partial_x f_s)^2 \cdot \partial_{yy} f_s - (\partial_y f_s)^2 \cdot \partial_{xx} f_s}{[(\partial_x f_s)^2 + (\partial_y f_s)^2]^2} \quad (17)$$

$$f(\varphi, \theta) = 1 - 15\varepsilon \cos(4(\varphi - \theta)) \quad (18)$$

where ε , φ and θ are the anisotropy parameter, angle between the horizontal axis and normal direction of the solid-liquid interface, and crystallographic orientation, respectively.

Third, the growth of each interface cell is tracked by a decentered square algorithm [37]. The dendrite begins to grow after the growth rate is determined. Crystal growth means that the solid

Table 2
The 2219 aluminum alloy microstructure simulation parameters.

Physical property	Value	Physical property	Value
Initial solute composition	6%	Diffusion coefficient (liquid)	$3.4 \times 10^{-9} \text{ m}^2 \cdot \text{s}^{-1}$
Solute partition coefficient	0.09825	Gibbs-Thomson coefficient	$1.0 \times 10^{-7} \text{ K} \cdot \text{m}$
Liquidus slope	$-2.717 \text{ K} \cdot \text{wt.}\%^{-1}$	Mesh size	$0.5 \mu\text{m}$
Diffusion coefficient (solid)	$1.48 \times 10^{-13} \text{ m}^2 \cdot \text{s}^{-1}$	Time step	$\frac{1}{5} \min(\frac{\Delta x}{v_{\text{max}}}, \frac{\Delta x^2}{D_L}, \frac{\Delta x^2}{D_S})$

fraction of CA cells increases. Therefore, the solid fraction of a CA cell $f_s|_{t+1}$ can be expressed as the solid fraction at the previous time $f_s|_t$ plus the increment of the solid fraction Δf_s . The corresponding equations are shown below [37]:

$$f_s|_{t+1} = f_s|_t + \Delta f_s \quad (19)$$

$$\Delta f_s = \text{GF} \frac{V_n \Delta t}{L_n} \quad (20)$$

$$L_n = \begin{cases} \Delta x \left(\frac{1}{|\cos \varphi|} + (1 - |\tan \varphi|) |\sin \varphi| \right) & |\cos \varphi| \geq |\sin \varphi| \\ \Delta x \left(\frac{1}{|\sin \varphi|} + (1 - |\cot \varphi|) |\cos \varphi| \right) & |\cos \varphi| < |\sin \varphi| \end{cases} \quad (21)$$

where Δt and Δx represent the time step and cell size, respectively. GF is a shape factor related to the state of neighboring cells. Their values and the microscopic properties of the material are summarized in Table 2.

3.3. Micro-macro coupling method

As mentioned above, solidification parameters are the bridge between macroscopic and microscopic simulations. The microscopic CA mesh is extremely fine, with a size on the order of micrometers ($0.5 \mu\text{m}$), while the macroscopic thermal flow field has a size on the order of millimeters (0.125 mm). Bilinear interpolation is the most adopted method to match meshes of different scales [19], and the principle of this method is illustrated in Fig. 1(c). The points $Q_{11}(x_1, y_1)$, $Q_{12}(x_1, y_2)$, $Q_{21}(x_2, y_1)$ and $Q_{22}(x_2, y_2)$ are the macroscopic grid nodes, and their temperature (T) values can be obtained from the macroscopic simulated results. The point $P(x, y)$ is the microscopic grid node, which is an unknown quantity to be solved. The bilinear interpolation method can be expressed as follows:

$$T(x, y) = \frac{\begin{bmatrix} x_2 - x & x - x_1 \end{bmatrix} \begin{bmatrix} T(Q_{11}) & T(Q_{12}) \\ T(Q_{21}) & T(Q_{22}) \end{bmatrix} \begin{bmatrix} y_2 - y \\ y - y_1 \end{bmatrix}}{(x_2 - x_1)(y_2 - y_1)} \quad (22)$$

4. Results and discussion

4.1. Macroscopic molten pool behavior

The temperature and flow distribution in the molten pool play an important role in weld formation and mechanical properties of welded joints [38]. EBW has the characteristic of a high energy density. When the electron beam acts on the base metal surface, the metal surface rapidly melts into a molten pool. With the continuous absorption of the energy of the electron beam, the volume of the molten pool increases, and a keyhole is formed as a result of the recoil pressure [39]. Fig. 2 shows the temperature field and the keyhole shape under different process parameters when the welding time is 100 ms. The weld penetration increases significantly with increasing beam current. The maximum velocity in the molten pool increases from 0.98 m/s in case A to 2.43 m/s in case D. A large beam current therefore leads to violent fluctuations in the molten pool, resulting in additional defects in the weld seam [40]. A series of bulges and contractions on the keyhole wall can be observed in Fig. 2(a2-d2), which is related to the absorption efficiency of the base metal to the electron beam [41]. An increasing

Table 3
Solidification parameters at the positions of interest.

	Thermal gradient (K/m)	Cooling rate (K/s)
Case A	4.02×10^5	7.85×10^4
Case B	5.50×10^5	5.96×10^4
Case C	7.31×10^5	5.03×10^4
Case D	1.12×10^6	4.08×10^4

amount of energy in the electron beam is absorbed by the bulges on the keyhole walls; therefore, a large thermal gradient is obtained around these regions, which results in an irregular distribution of the recoil pressure and surface tension. Eventually, this may cause the keyhole to collapse.

To validate the mathematical models developed in this paper, the simulated and experimental weld profiles obtained under the different welding parameters are compared, as shown in Fig. 2(a3-d3) and (a4-d4). The predicted weld morphologies are in good agreement with the experimental morphologies. This indicates that the model established in this study is reasonable. Specific dimension information, such as the penetration depth, is summarized in Fig. 3. A slight difference is observed between the simulations and experiments. The reason for this phenomenon may be the error induced by manual focus adjustment, which can lead to drastic variations in the weld depth [42].

The stability of keyholes is another factor that can be used to evaluate the quality of welded joints. A stable keyhole can reduce the possibility of spiking defects in the weld seam [40]. Fig. 4 plots the evolution of the keyhole depth under different process parameters. Apparently, the keyhole depth and deep penetration velocity increase with increasing beam current. Moreover, the keyhole also requires additional time to achieve a steady state. For example, the keyhole depth is dynamically stable at approximately 70 ms in case D, while it only takes 30 ms to achieve stability in case A. When the keyhole is stable, the average keyhole depth is 8.62 mm, 9.73 mm, 11.61 mm and 13.58 mm for cases A, B, C, and D, respectively. In addition, the keyhole oscillation is obvious when there is a large beam current. Therefore, it is demonstrated again that the fluctuation of the molten pool is more violent with a large beam current. As mentioned above, the bulges on the keyhole wall reduce the possibility of the electron beam energy reaching the keyhole bottom. The deeper the keyhole is, the lower the probability and the more unstable the keyhole. Therefore, it can be concluded that a larger beam current is more inclined to generate spiking defects, despite the increase in the penetration depth.

To compare the microstructure of the weld seam under different welding parameters, the same positions on the weld centerline, which is 1 mm away from the base metal surface, are chosen for microscopic simulation and experimental observation, as shown in Fig. 1(d). The solidification parameters, the thermal gradient and cooling rate at these positions are extracted, as presented in Table 3. The thermal gradient, which is the ratio of the temperature difference between adjacent isothermal surfaces and their vertical distance, increases with increasing beam current. This is due to an increase in the welding heat input, which increases the density of the isothermal surfaces. The cooling rate, the rate at which the temperature decreases with time, decreases with in-

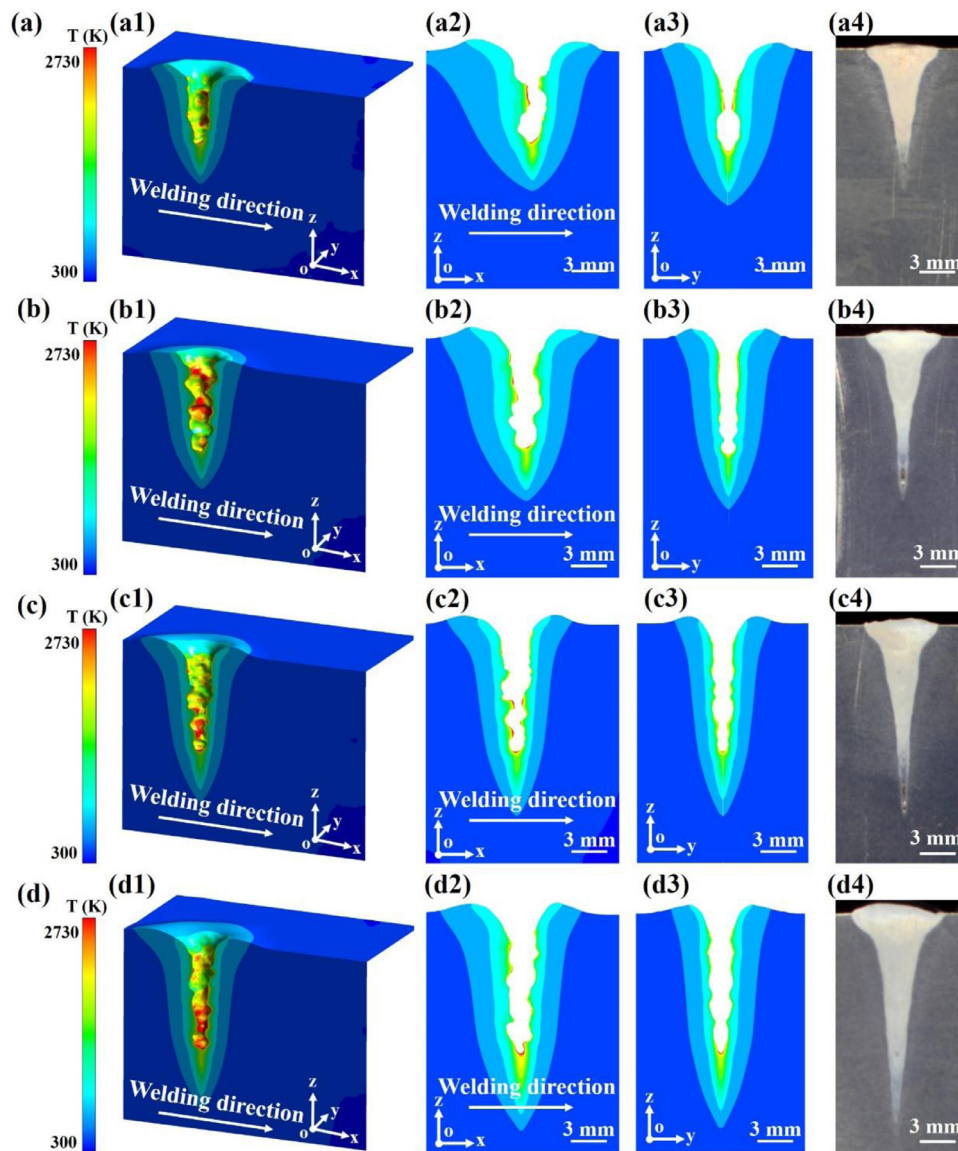


Fig. 2. The molten pool behavior and keyhole shape under different process parameters when the welding time is 100 ms: (a) Case A, 40 mA; (b) Case B, 50 mA; (c) Case C, 55 mA; and (d) Case D, 60 mA. In this figure, (a1-d1), (a2-d2) and (a3-d3) are the results presented in a 3D view, longitudinal-sectional side view and cross-sectional side view. (a4-d4) are the weld morphologies under different process parameters.

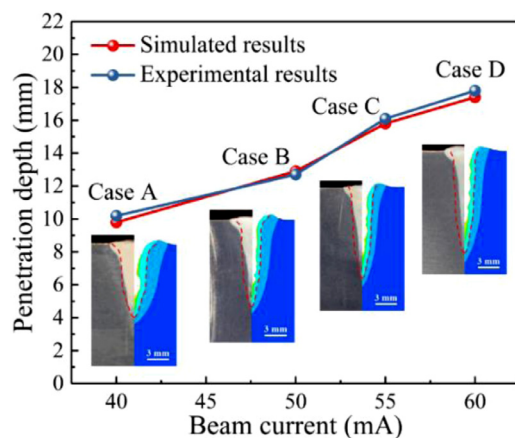


Fig. 3. Comparison of the simulated and experimental weld profiles under different welding parameters.

creasing beam current. This is because a larger heat input prolongs the residence time of the molten pool in the high-temperature region. The variation of these solidification parameters effects the microstructure in the weld seam.

4.2. Microstructure evolution

To investigate the characteristics of the grain growth under such thermal conditions of the EBW process, we determined the microstructure evolution mechanism before comparing the microstructure characteristics of different welding parameters. Although the temperature field exists in the form of a three-dimensional spatial distribution, we can simplify the microstructure simulation area of the longitudinal section into a two-dimensional space due to the direction of heat flow. The description of this simplified method can be found in the Supplementary materials. As mentioned above, a computational domain with a size of $130 \mu\text{m} \times 130 \mu\text{m}$ and a grid resolution of $0.5 \mu\text{m} \times 0.5 \mu\text{m}$ is set to simulate the microstructure evolution at the position of interest, as shown in Fig. 1(d). The simulations are computed by

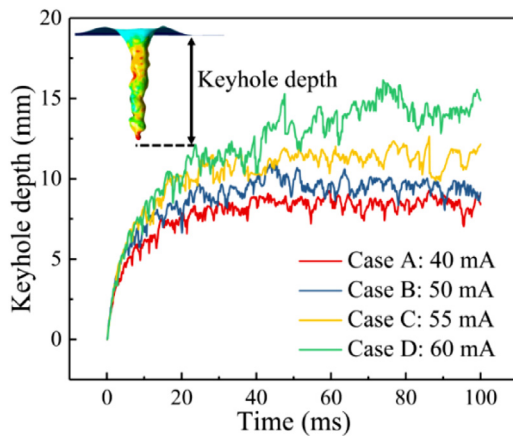


Fig. 4. The keyhole depth evolution under different process parameters.

a fixed time step of 10^{-6} s. Each grid in the computational domain has a fixed cooling rate and a vertical thermal gradient. The values of these solidification parameters are extracted from the macroscopic simulations. At the beginning of the calculation, the bottom of the computational domain is set as a solid-liquid interface, and its temperature is 930 K. Fig. 5(a-e) presents the evolution of the solute concentrations during the solidification process of the EBW molten pool (case D). All grains are given a color to represent their morphologies on a red-green-blue (RGB) color scale. The method used to color each grain involved taking a random number from the R, G and B value range from 0 to 255 and mixing these three colors, as presented in Fig. 5(f-j).

A crystal in the computational domain is chosen as an example to analyze the formation mechanism of the crystal, as shown in the enlarged insert in Fig. 5(a-c). In the initial period of solidification, nucleation occurs at the bottom of the computational domain, as shown in Fig. 5(a) and (f). The equiaxed crystals have sufficient space for growth and do not interfere with each other. The crystals form four sharp corners and appear to have a symmetrical structure. Moreover, solute atoms are discharged from the solid phase, and a thin solute enrichment layer can be observed around the crystals. As crystallization proceeds, the solute enrichment layer becomes increases in thickness, as seen in Fig. 5(b) and (g). When crystals with different crystal orientations grow to encounter and collide, the growth mechanism changes from unconstrained growth to competitive growth, as shown in Fig. 5(c) and (h). Due to the direction of the thermal gradient, nucleation in the calculation domain is not simultaneous, but from bottom to top, as shown in Fig. 5(a-d) and (f-i). This phenomenon is consistent with the simulation results [43] and experimental observations [9]. The crystal formation experiences nucleation, growth and competitive growth. When the studied region is completely solidified, the concentration distribution of the microstructures is determined and shown in Fig. 5(e), and the solidified microstructures is shown in Fig. 5(j).

Visually, most of the crystal morphology is asymmetric after solidification, as marked by the red circles in Fig. 5(e). This phenomenon is consistent with the results in Ref [44]. During the stage of competition growth, the thermal gradient has a selective effect on the grain growth [45]. The grains with a favorable orientation (FO) grow faster than those with an unfavorable orientation (UO). One of the reasons is that the angle between the growth direction of the FO grains and the direction of the thermal gradient is smaller than that for the UO grains. The FO grains can obtain increased undercooling around the side branches, thus increasing the dendrite growth rate. After UO grains survive competitive growth, the forward growth of the lateral branches of UO grains is hindered

Table 4

Chemical composition on the grain boundary (at.%). The measured positions are indicated by the black crosses in the grain boundary in Fig. 6(b).

Grain boundary	Al		Cu	
	Simulated	Experimental	Simulated	Experimental
P ₁	88.23	88.61	11.77	11.39
P ₂	89.40	89.88	10.60	10.12
P ₃	88.43	86.35	11.57	13.65
P ₄	88.92	87.37	11.08	12.63
P ₅	90.52	92.31	9.48	7.69
P ₆	92.61	91.44	7.39	8.56
Average value	89.69	89.33	10.31	10.67

Table 5

Chemical composition inside the grain (at.%). The measured positions are indicated by the white crosses inside the grain in Fig. 6(b).

Grain interior	Al		Cu	
	Simulated	Experimental	Simulated	Experimental
P ₇	96.94	97.07	3.06	2.93
P ₈	96.83	97.45	3.17	2.56
P ₉	98.16	98.26	1.84	1.74
P ₁₀	97.98	97.15	2.02	2.85
P ₁₁	96.74	97.83	3.26	2.17
P ₁₂	98.03	98.11	1.97	1.89
Average value	97.45	97.64	2.55	2.36

so that they can only grow in the axial direction. After a while, the grain boundaries are formed between the crystals. Many solute atoms are distributed on the grain boundaries, which may lead to segregation.

The concentration distribution plays an important role in the microstructure evolution. It not only determines the growth kinetics of the solidification interface (as illustrated in Eq. (14)) but also affects the segregation, which has an adverse effect on the welded joint [16]. It can be observed from Fig. 5(e) that the concentration of Cu at the grain boundaries is higher than that inside the grains. To verify the element distribution of the simulated microstructure, high-angle annular dark-field scanning transmission electron microscopy (HAADF-STEM) surface scanning of the elements around a grain is conducted, as presented in Fig. 6(b-d). Moreover, a grain is selected randomly from the simulated results to study the element distribution, as seen in Fig. 6(e) and (e). The experimental results indicate that the elements are unevenly distributed around the grain boundaries. Cu is enriched on the grain boundaries, and there is a higher content than that inside the grains. The simulated results and experimental observations present the same element distribution pattern, which can also illustrate the rationality of the numerical model.

Fig. 6(b) shows a grain boundary and the interior of a grain, which are selected for elemental analysis. The same number of positions from the simulated grain boundary and inside the grain are selected, and the results are listed in Table 4 and Table 5. According to the EDS results, the concentration of Cu on the grain boundary is significantly higher than the initial concentration (6%), and the average value is 10.67%. The average concentration of Cu inside the grain reaches 2.36%. These phenomena are related to the migration of solute atoms during the microstructure evolution [37]. Accordingly, the simulated average concentrations of Cu on the grain boundary and inside the grain are 10.31% and 2.55%, respectively. These results illustrate the accuracy of the model.

4.3. Simulated results of the microstructure with different process parameters

Increasing the beam current increases the depth-to-width ratio of the weld, but an excessively high beam current may cause an

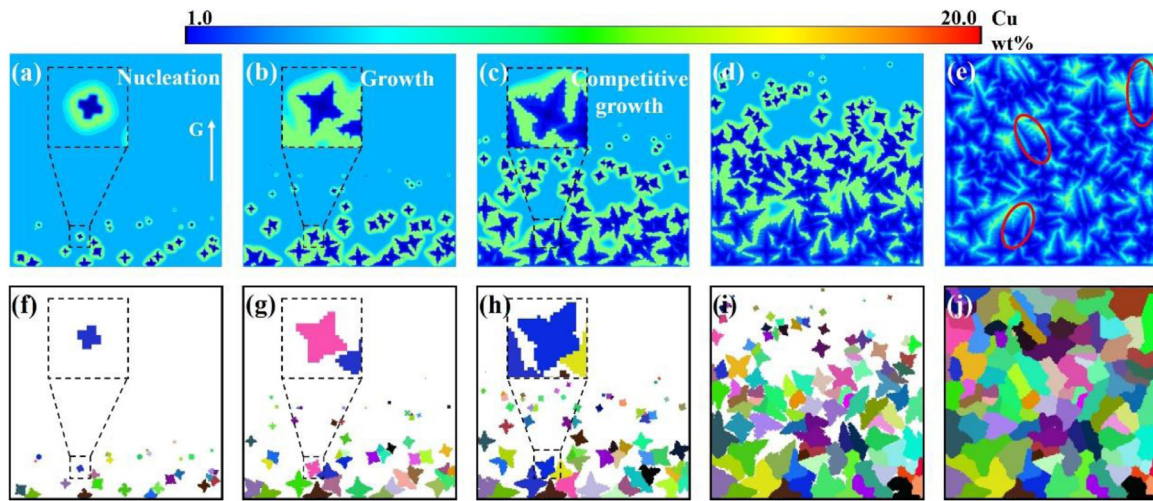


Fig. 5. Evolution of the solute concentrations and grain structures during the solidification process of the EBW molten pool (Case D): (a, f) 1 ms, (b, g) 1.5 ms, (c, h) 2.1 ms, (d, i) 3 ms, and (e, j) 8 ms.

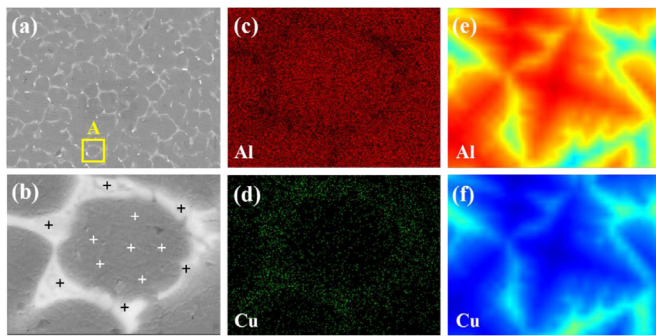


Fig. 6. Microstructure and regional elemental distribution: (a) SEM results, (b) enlarged microstructure of area A, (c, d) HAADF-STEM surface scanning results and (e-f) simulated elemental distribution results.

unstable molten pool, as shown in Fig. 4. To investigate the effect of the beam current on the grain structure, the microstructure evolution is simulated with different process parameters. Fig. 7 shows a comparison of the simulated grain structure with the experimental EBSD results under different welding parameters. It can be seen in Fig. 7(a) that the weld center after EBW is composed of a large number of equiaxed grains. In case A, the number of equiaxed grains in the center of the weld is large, and the size is small (Fig. 7 a1). In case D, the number of crystals decreased, and the size increased obviously (Fig. 7 a4). There are two main reasons for this variation: (1) with increasing beam current, the energy input increases and the cooling rate of the solidification interface decreases, resulting in a decreased nucleation undercooling and number of grains and (2) the grains have more time to grow due to the decrease in the cooling rate. From Fig. 7(a) and (b), the simulated results show fairly similar grain distribution patterns to those obtained with EBSD. The grain size distributions are counted, as illustrated in Fig. 7(c). The simulated grain size distribution shows a trend of first increasing and then decreasing with different process parameters. However, the grain size distribution determined by the experiments is more disordered. This may be because the thermal gradient and cooling rate of the studied region are not fixed values but change constantly in the actual solidification field.

Fine grain strengthening is one of the strengthening mechanisms of weld seams [20]. Fig. 8 shows the simulated and experimental average grain sizes. For cases A to D, the simulated average grain sizes are 5.13 μm , 6.73 μm , 8.46 μm and 9.76 μm , re-

spectively, while the experimental average grain sizes are 4.54 μm , 6.87 μm , 7.68 μm and 8.71 μm , respectively. The relative error is approximately 10%, which may be caused by the deviation of the thermal gradient and cooling rate mentioned above. The simulated and experimental results show that the grain size tends to increase with increasing beam current. Therefore, we can conclude that the smaller the electron beam current is, the smaller the grain size is, and the higher the strength of the weld seam may be.

The solidification process of the molten pool during EBW involves rapid cooling. The ejected solute atoms have insufficient time to diffuse, resulting in an uneven distribution of chemical composition in the local solidification region, as shown by the EDS results in Fig. 9(a-d). This segregation phenomenon, which leads to a solidification defect related to the hot crack, is unfavorable to the performance of welded joints [46]. Therefore, it is necessary to study the influence of different welding parameters on the segregation. It is well known that the degree of segregation is related to the solidification time. Fig. 9(e) shows the evolution of the solid fraction with solidification time under different process parameters. It can be seen that the higher the heat input is, the longer the solidification time. Fig. 9(f) shows the segregation evolutions with solid fraction f_s under different process parameters. In this study, the segregation ratio is defined as follows [47]:

$$SR = \frac{\sum_{i,j} C_{i,j}^l (1 - f_s^{i,j})}{\sum_{i,j} C_0 (1 - f_s^{i,j})} \quad (23)$$

The simulated results are evaluated by comparing them with the Scheil analytical model, which is not sensitive to the thermal gradient and cooling rate because it assumes that the solid phase does not diffuse, and the liquid phase is uniformly mixed. Since a series of assumptions are made in the derivation, the Scheil equation is not applicable with f_s close to 1 and is only an f_s function. With the nucleation and growth of crystals, an increasing number of solute atoms are discharged into the liquid phase, resulting in a large solidification region of low concentration and a high concentration in the melt, as shown in Fig. 5(a-e). Therefore, the segregation ratio increases during solidification. The simulated segregation ratio evolution trends are consistent with the Scheil model. The solidification times are 3.33 ms, 4.27 ms, 5.83 ms and 7.70 ms for case A, case B, case C and case D, respectively. At $f_s=0.8$, the segregation ratios are 4.36, 4.02, 3.69 and 3.23 for case A, case B, case C and case D, respectively. It can be inferred reasonably that the shorter the solidification time is, the more severe the segrega-

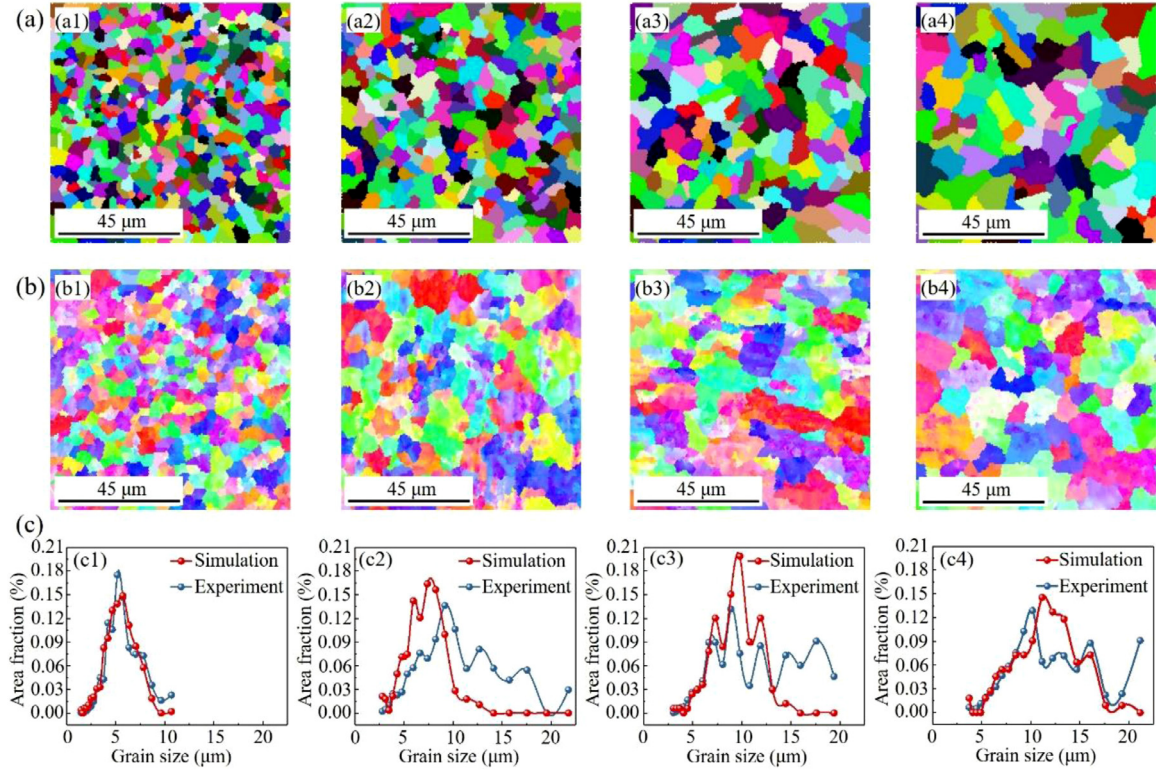


Fig. 7. Grain structure of the position in Fig. 1(d): (a) simulated results, (b) experimental EBSD results and (c) grain size distributions from the simulations and experiments. In this figure, (a1-c1), (a2-c2), (a3-c3) and (a4-c4) are the results for case A, case B, case C and case D, respectively.

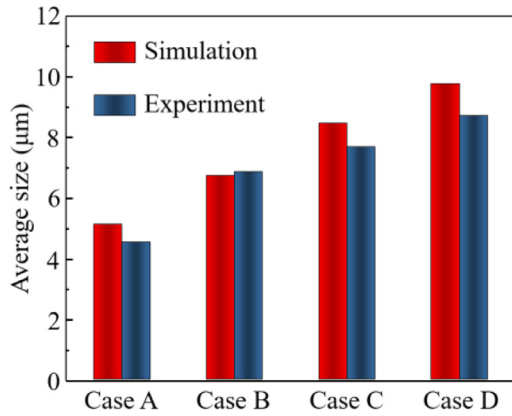


Fig. 8. Simulated and experimental average grain size.

tion in the weld seam. Therefore, a weld seam with a higher beam current may have better crack resistance.

4.4. Effect of the solidification parameters on the grain size

Four fixed cooling rates are adopted to study the effect of the thermal gradient on the grain size. The values of the cooling rate are obtained from Table 3. The values of the thermal gradient are extended into a matrix and range from 4.02×10^5 to 1.12×10^6 K/m. The microstructure evolutions are simulated with these extended solidification conditions. The average grain size is determined, as plotted in Fig. 10(a). At $CR=4.08 \times 10^4$, the grain size increases slowly with increasing thermal gradient. At $CR=12.9 \times 10^4$, this trend is not obvious. By fitting the curves for the grain size and thermal gradient, the correlation coefficients are 0.506, 0.496, 0.449 and 0.175 for cooling rates of 4.08×10^4 K/s, 5.03×10^4

K/s, 7.96×10^4 K/s and 12.9×10^4 K/s, respectively. The correlation coefficient decreases with increasing cooling rate. Even at a low cooling rate (i.e., $CR=4.08 \times 10^4$ K/s), the grain size is weakly correlated with the thermal gradient. At a high cooling rate (i.e., $CR=12.9 \times 10^4$ K/s), it can be said that the crystal size has little correlation with the thermal gradient. Therefore, it can be inferred that the thermal gradient has little effect on the grain size, which is consistent with the conclusion in reference [17].

To understand the effect of the cooling rate on the grain size, the microstructure evolutions are simulated by a series of solidification conditions. A fixed thermal gradient of 7.01×10^5 K/m is set in the whole computational domain, which represents the mean value calculated from Table 3. The values of the cooling rate are extended into a matrix and range from 4.08×10^4 to 1.29×10^5 K/s. The relationship between the cooling rate and grain size is presented in Fig. 10(b). It is interesting to note that the grain size versus cooling rate is much steeper than any single curve with a constant thermal gradient (Fig. 10a). The grain size and cooling rate have a power-law relationship of $d \propto CR^{-0.55}$ by fitting all extended data, which is similar to $d \propto CR^{-0.5}$ calculated by Easton et al. [48]. It can be considered that during the rapid cooling process of EBW, the size of the local microstructure mainly depends on the cooling rate, and the effect of the thermal gradient can be ignored, as described in reference [49].

4.5. Effect of the solidification time on the segregation ratio

The segregation phenomenon describes the uniformity of the element concentration in the weld microstructure. When the diffusion rate of ejected solute atoms during solidification is determined, the degree of segregation mainly depends on the solidification time. Fig. 11(a) shows the solidification time of a local region with different thermal gradients and cooling rates. The solidification time is gradually prolonged with increasing thermal gradient

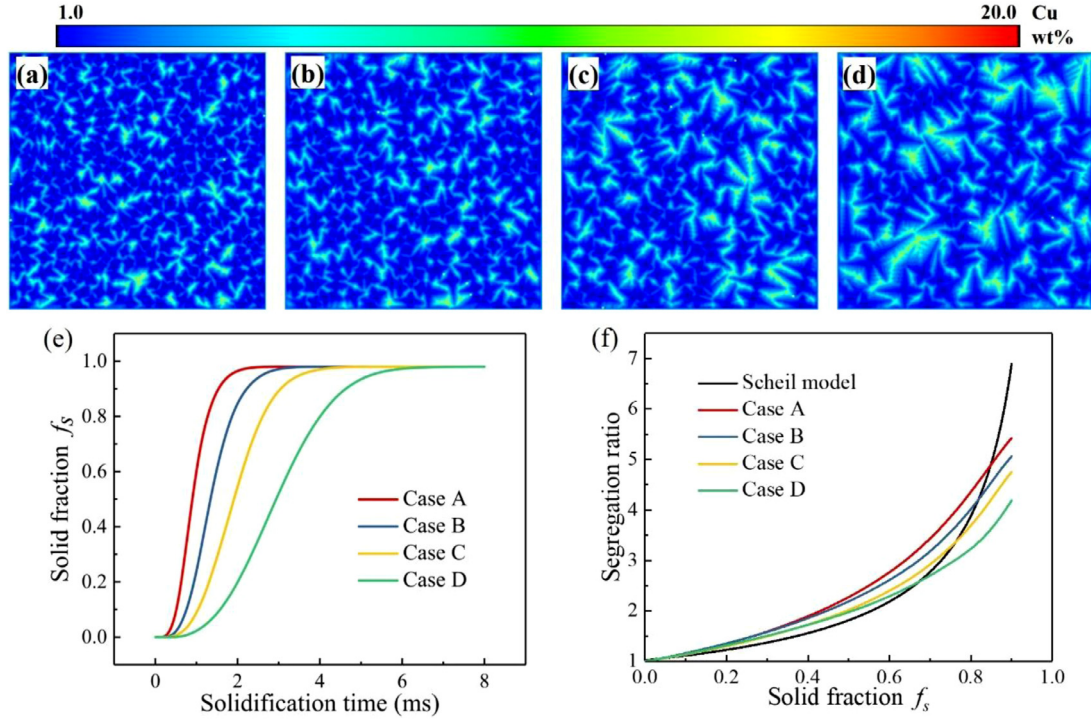


Fig. 9. (a-d) Concentration distribution of Cu for case A, case B, case C and case D, respectively. (e) Evolution of the solid fraction with solidification time and (f) segregation evolution with f_s under different process parameters.

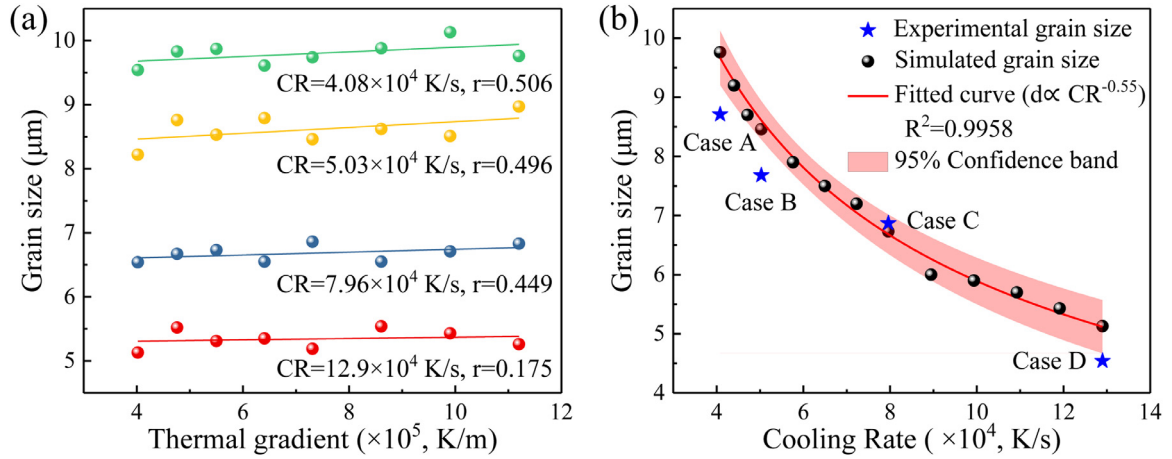


Fig. 10. The effect of (a) the thermal gradient and (b) the cooling rate on the grain size.

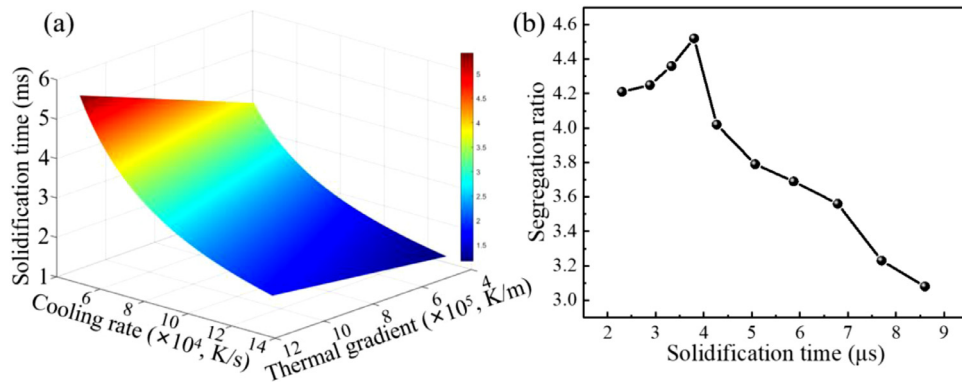


Fig. 11. The effect of (a) thermal gradient and (b) cooling rate on the segregation ratio. The segregation ratio is calculated at $f_s=0.8$.

and decreasing cooling rate. At $f_s=0.8$, the segregation ratios under different solidification times are calculated, as plotted in Fig. 11(b). As observed, the segregation ratio increases when the solidification time is reduced. This is because the shorter the migration time of the solute atoms is, the more inhomogeneous the element diffusion, resulting in an increased segregation ratio. However, continuing to reduce the solidification time decreases the segregation ratio slightly. When the solidification time is decreased, more equiaxed grains grow faster, and the low concentration region is expanded [47]. Because there is a small grain size and elimination of solute atoms being discharged from the solid phase, the segregation ratio is reduced. As observed in Fig. 11(a), the effect of the cooling rate on the solidification time is greater than that of the thermal gradient. Therefore, it can be reasonably inferred that the effect of the cooling rate on segregation is greater, which was also proven by Wang's study [47].

5. Conclusion and outlook

In this paper, a multi-scale model is established to simulate the whole welding process. On the macroscopic scale, the molten pool dynamic is visualized by a thermal-fluid model, and the keyhole stability is evaluated. On the microscopic scale, the microstructure evolution process is investigated by a CA model. The simulated results are verified by experiments. Some basic conclusions can be summarized as follows:

- (1) With an increase in the beam current, the molten pool becomes unstable, and the keyhole oscillation increases in severity, which means that a larger beam current is more inclined to generate spiking defects. The thermal gradient in the molten pool increases, and the cooling rate decreases. The simulated weld seam size and morphology match the experimental measurements.
- (2) During the formation of the microstructure, each grain experiences nucleation, growth and competitive growth processes. After solidification, the concentration of Cu on the grain boundary is significantly higher than the initial concentration, while the result is the opposite inside the grain.
- (3) With increasing beam current, the grain size tends to increase, and the segregation ratio decreases. The weld seam with a smaller electron beam current may have higher strength but lower crack resistance. The simulated microstructure morphologies show similar patterns to those obtained with EBSD. The simulated element content is verified by EDS results.
- (4) The grain size is weakly correlated with the thermal gradient. At higher cooling rates, the effect of thermal gradient on grain size can be ignored. The grain size mainly depends on the cooling rate and has a power-law relationship of $d \propto CR^{-0.55}$.
- (5) The degree of segregation is affected by the solidification time. With an increase in the thermal gradient and the decrease in the cooling rate, the solidification time increases. With a decrease in solidification time, the segregation ratio first increases and then decreases slowly.

In summary, the electron beam current should be neither too small nor too large. Too small a beam current may decrease the penetration depth and reduce the crack resistance of the weld seam. Too large a beam current may lead to more defects and reduce weld strength. The simulated results demonstrate the ability and accuracy of the multi-scale model to reproduce the evolution of melting and solidification processes in the entire weld. This multi-scale model can be adopted to predict weld formation and the microstructure. This research lays a good foundation for optimizing the welding process parameters.

In the future, it will be meaningful to perform 3D microstructure simulations. Although many phenomena/mechanisms found in this work are well consistent with the analytical theories and the experimental results, the microstructure evolution in 3D is more complicated. For example, the grain growth direction and solute diffusion channel are more pluralistic, and competitive growth between different dendrites is more complex in 3D space. On the other hand, the calculation of the whole weld seam region (not the local region) is also very interesting. In this way, we can analyze the microstructure (grain size, segregation and so on) at the different positions in the weld seam, which provides a more sufficient basis for improving the welding process.

Data Availability

The raw/processed data required to reproduce these findings cannot be shared at this time, as the data are part of an ongoing study.

Declaration of Competing Interest

The authors declared that there is no conflict of interest.

CRediT authorship contribution statement

Ziyou Yang: Writing – review & editing, Investigation, Conceptualization, Software. **Kangning Jin:** Software, Methodology. **Hui Fang:** Writing – review & editing, Visualization. **Jingshan He:** Supervision, Conceptualization.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:[10.1016/j.ijheatmasstransfer.2021.121156](https://doi.org/10.1016/j.ijheatmasstransfer.2021.121156).

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